

Sugar Bombs and Shelf Space:
A Cluster Analysis of Cereal Nutrition and Marketing Tricks

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Motivation for the Dataset

Breakfast cereals come in all sorts of varieties, from sugar-packed children’s cereal to healthier bran cereals. Having recently been told about a trick supermarkets play as a marketing tactic, I decided to investigate if we can split cereals into nutritional clusters and whether the sugary ones are placed at a child’s eye level with frequency beyond reason.

Data & Methods

The dataset contains 77 breakfast cereals with per-serving nutrition. After cleaning up the data by removing incomplete rows, 74 cereals remain, described by eight nutrients: calories, protein, fat, sodium, fibre, carbohydrates, sugars and potassium.

The nutrients were normalised per ounce, as opposed to the arbitrarily chosen serving sizes, and then scaled appropriately using R’s built-in function. Shelf height and the composite rating were held out of the clustering: shelf height because I planned on testing if one of the clusters and a shelf height were correlated, so including shelf height would be circular, and rating was based on the nutrients which were already being included.

Definitions

- K-means:** Partitional clustering algorithm into k groups around their means.
- Silhouette:** A measure of how well-separated and compact clusters are. (−1 to 1).
- Hopkins Statistic:** A test to see if data is non-random. (~0.5 random, ~1 clustered).
- ARI:** A measure of agreement between two clusterings. (1 ⇒ identical, 0 ⇒ chance).
- PAM:** Partitional clustering algorithm into k groups around medoids.
- Agglomerative Clustering:** Hierarchical clustering algorithm where you start with n clusters and repeatedly merge the nearest points or clusters.

Is there structure?

The Hopkins statistic¹ (dimension-corrected; Wright, 2022) of 0.998 confirmed the data is not “purely random white noise” and it has real shape. However, this only tests non-randomness, not whether the groups are well separated.

How many clusters?

Both the elbow and silhouette methods indicated a weak, continuous structure with no obvious optimal k . Silhouette widths peaked at a weak 0.3, indicating high overlap rather than obvious distinct clusters, which we’d expect from the continuum in cereal nutrients. This is likely due to a low b(i) in the silhouette. It’s important to note this doesn’t make the dataset ‘bad’ for clustering, rather it is open to multiple interpretations.

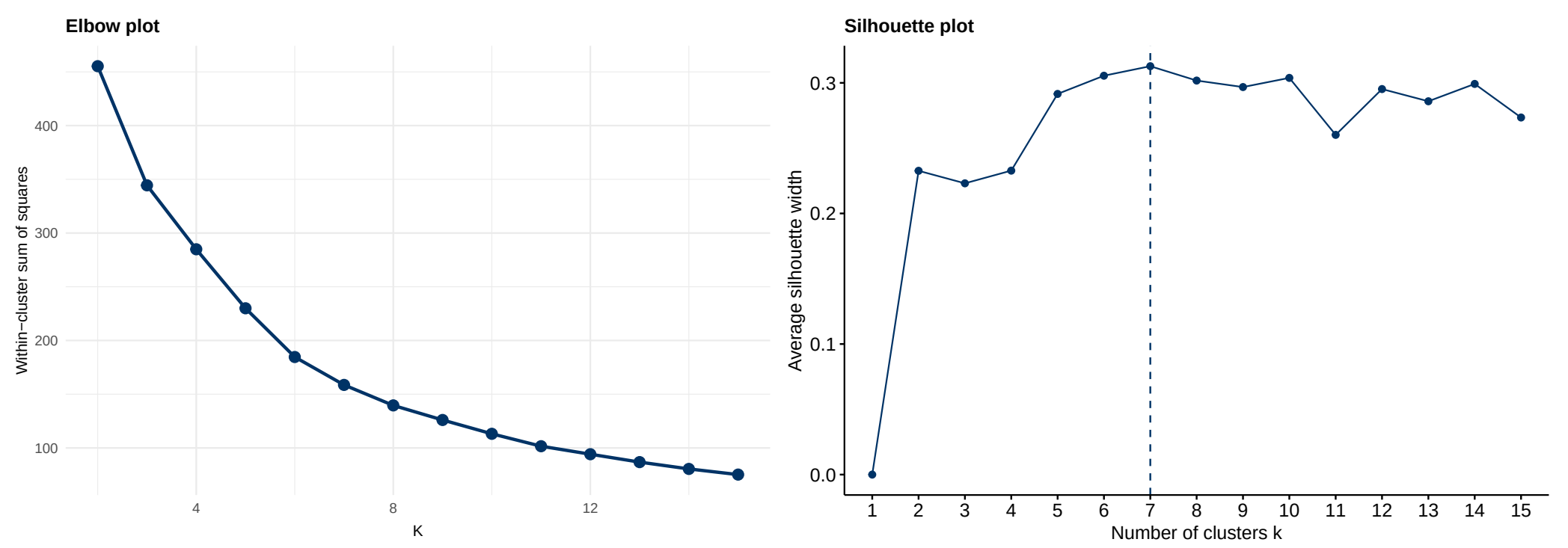


Figure 1: Elbow and silhouette plots: no sharp elbow and a flat plateau from $k = 5$ to 10

How can we best select k ?

The task of selecting k becomes challenging, as there isn’t a clear peak or elbow on either plot. The silhouettes give a range which may give rise to decent clustering ($k = 5$ to 10). Hence, we should select k based on interpretability and whether the clusters are recognisable categories of cereal. Manually inspecting the cases above, $k = 5$ yielded the most recognisable groups, with larger k just splitting up categories into near-identical ones.

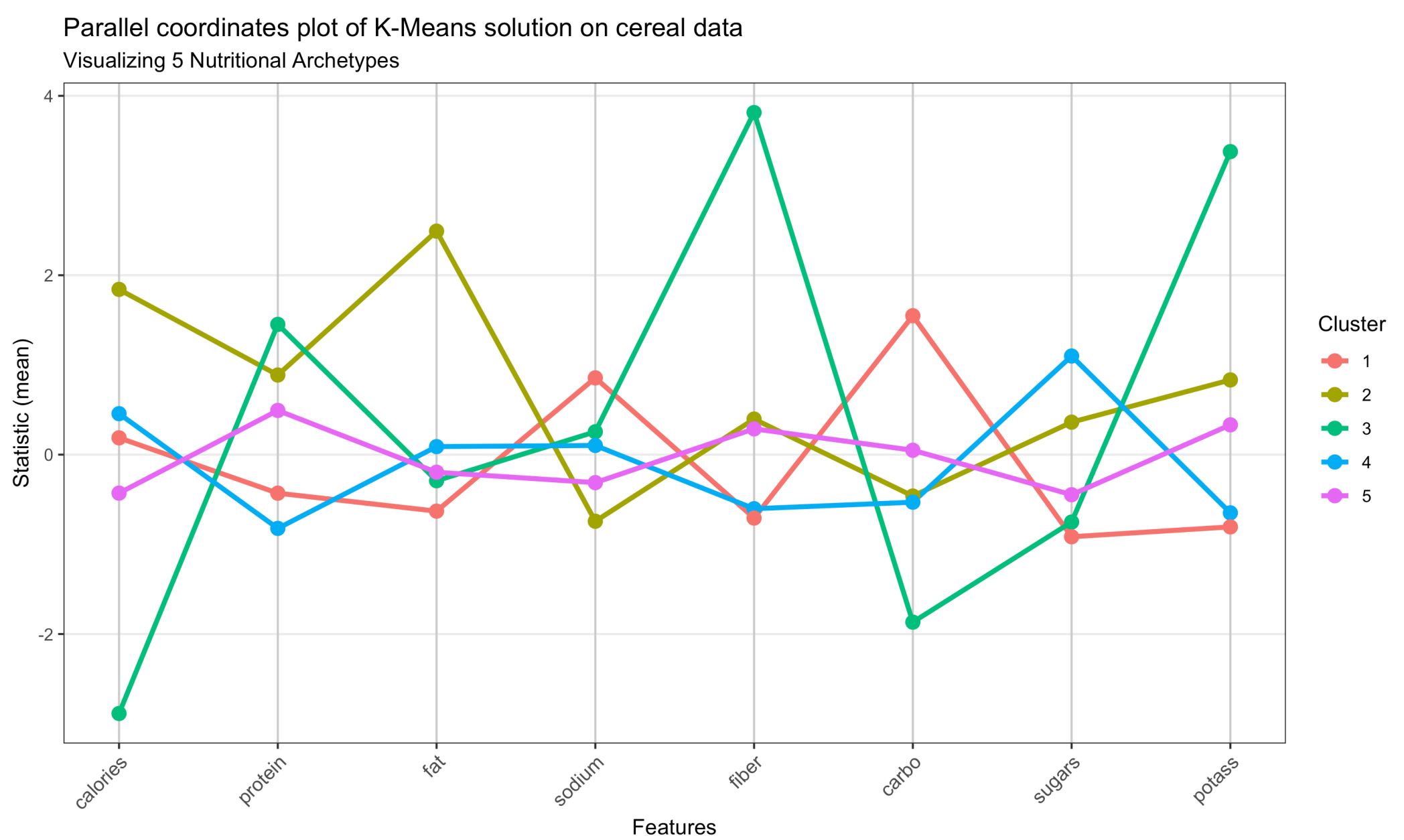


Figure 2: A parallel coordinates plot of the clustering created by K-means, with $k = 5$ and the euclidean distance metric.

Five nutritional archetypes

- Cluster 1:** Plain rice or corn flakes
- Cluster 2:** Granola or Muesli
- Cluster 3:** High-fibre brans
- Cluster 4:** Sugar bombs
- Cluster 5:** Mainstream balanced

Cross-method validation

Three algorithms, k-means, agglomerative with ward linkage and k-medoids (PAM), were compared using the Adjusted Rand Index. They agreed moderately (ARI 0.38–0.56), with all three recovering the same cluster cores (the clear sugar bombs and brans), while the intermediate cereals are sorted into different clusters, reinforcing that we are dealing with a continuum. Strong agreement would imply highly distinct clusters, but our partial agreement suggests the boundaries between intermediate clusters are blurry.

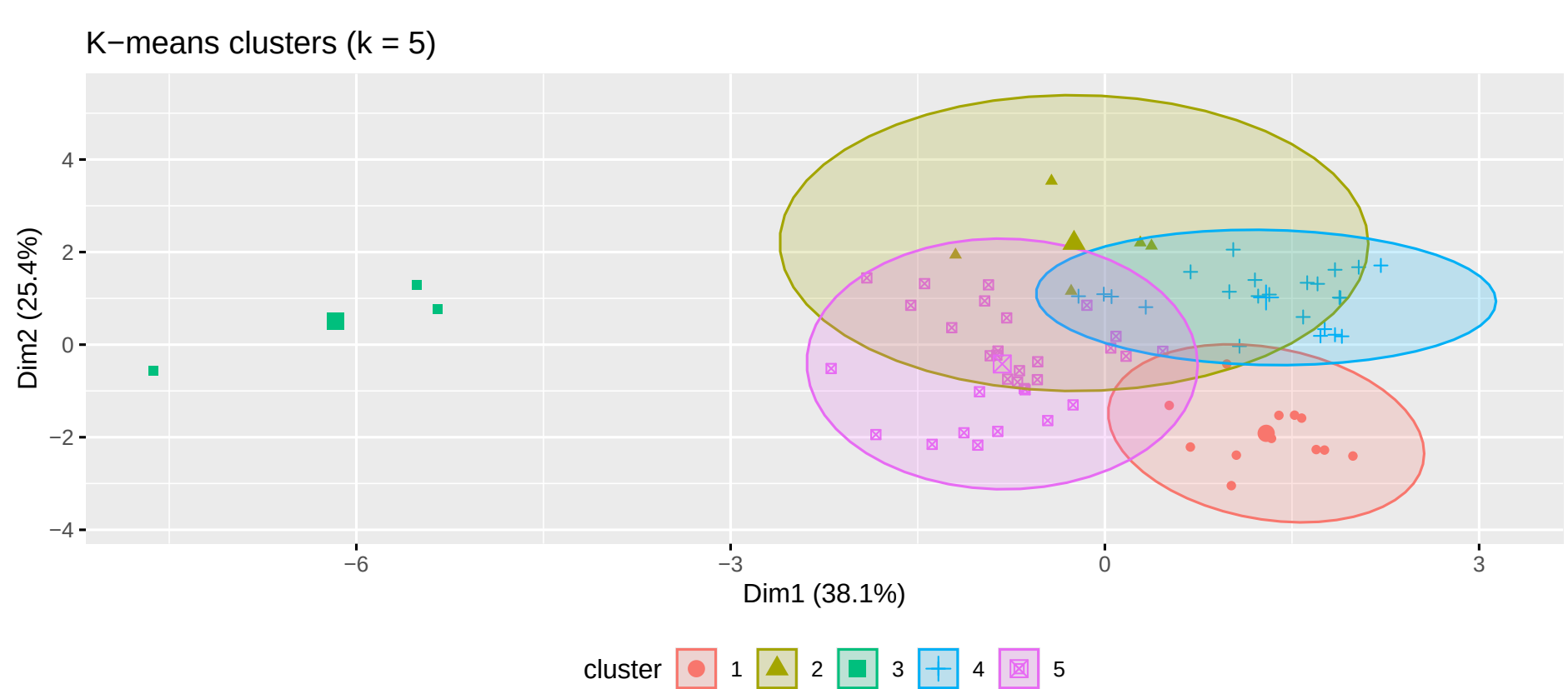


Figure 3: The five clusters on the first two principal components². Cores separate but boundaries overlap, reflecting the continuum.

Do sugar bombs target kids?

Having noticed high-sugar cereals are placed at a child’s eye level, I tested whether the sugar-bomb cluster is over-represented on shelf 2 (the middle, child-height shelf), using a binomial hypothesis test.

$$p = 0.0062$$

$$H_0: p = 1/3$$

$$H_1: p > 1/3$$

Under the null hypothesis, we found that $p = 0.0062$, which gives us sufficient evidence to reject H_0 at the 1% significance level. This suggests that the sugary cereals may be over-represented in shelf 2.

Sugary cereals are disproportionately placed at a child’s eye level

Conclusion

Cereals form a nutritional continuum rather than distinct clusters, yet a five-cluster segmentation captures recognisable categories that hold well across methods. The sugary cluster is also disproportionately placed at shelf 2, consistent with the marketing hypothesis.

Limitations

- Weak, continuous structure. The intermediate clusters have overlap, despite the 2 extremes being consistent.
- Small sample ($n \approx 74$)
- k chosen on interpretability rather than a statistical optimum.
- The shelf association is correlational, which does not imply causation.

References

- Wright, M. N. (2022). Will the Real Hopkins Statistic Please Stand Up? The R Journal.
- Holbert, C.F. (n.d.) Clustering on Principal Component Analysis. Available at: <https://www.cfholbert.com/blog/cluster-pca/>
- Use of Generative AI for formatting and LaTeX

Affiliations

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